1.2 Process $B \to C$

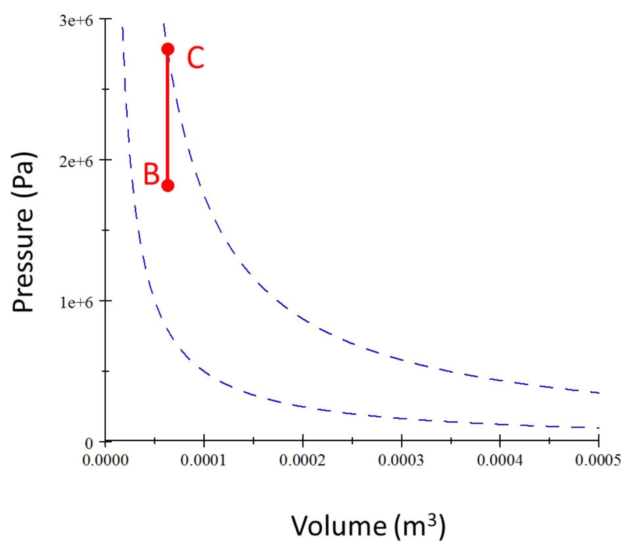
Assignment"

Fill in the table for process $B \to C$

ANSWER:

State	**T** (K)	**P** (kPa)	$\cancel{V}$ (m^3)	**n** (mol)	$\mathbf{E}_{int}$ (J)	–
B	673.15	1837.9	6.25×10^{-5}	2.0525×10^{-2}	287.17	–
C	1023			2.0525×10^{-2}		–
Process	**Q** (J)	$\mathbf{w}_{int}$ (J)	$\mathbf{\Delta E}_{int}$ (J)	$\mathbf{w}_{eng}$ (J)	$\mathbf{Q}_h$	Q_c
BC						

Process $B \to C$ is an isochoric process.



This is where the spark plug ignites the air-fuel mixture created by the fuel injectors. A great deal of energy is released in a hurry. The temperature jumps up and the pressure does too, but the volume is the same because this process happens in a flash–literally!

So this must be an isochoric process. We expect $w_{BC} = 0$, so $\Delta E_{BC} = Q_{BC}$. The pressure increases, so we expect Q_{BC} to be positive. That means this energy is entering our engine. This must be part of Q_h! We know the temperature at

C and we know the volume $\cancel{V}_C = \cancel{V}_B$ so we can find the final pressure

$$\begin{aligned} P_C &= \frac{nRT_C}{\cancel{V}_C} \\ &= \frac{\left(2.052\,5 \times 10^{-2}\right)\left(8.314\frac{\mathrm{J}}{\mathrm{mol\,K}}\right)(1023)}{6.25 \times 10^{-5}\,\mathrm{m}^3} \\ &= 2.793\,1 \times 10^6\,\mathrm{Pa} \\ &= 2793.1\,\mathrm{kPa} \end{aligned}$$

We can find the internal energy at C now

$$\begin{aligned} E_C &= \frac{5}{2}nRT_C \\ &= \frac{5}{2}\left(2.052\,5 \times 10^{-2}\,\mathrm{mol}\right)\left(8.314\frac{\mathrm{J}}{\mathrm{mol\,K}}\right)(1023\,\mathrm{K}) \\ &= 436.42\,\mathrm{J} \end{aligned}$$

We can use

$$\begin{aligned} Q &= nC_V\Delta T \\ &= n\frac{5}{2}R\left(T_C - T_B\right) \end{aligned}$$

For the process BC to find the energy transferred by heat

$$\begin{aligned} Q_{BC} &= \left(2.052\,5 \times 10^{-2}\,\mathrm{mol}\right)\frac{5}{2}\left(8.314\frac{\mathrm{J}}{\mathrm{mol\,K}}\right)(1023\,\mathrm{K} - 673.15\,\mathrm{K}) \\ &= 149.25\,\mathrm{J} \end{aligned}$$

We already mentioned that $Q_{BC} = \Delta E_{int_{BC}}$ and $w_{BC} = 0$, so we can complete our table of state variables and process variables for this process.

State	**T** (K)	**P** (kPa)	$\cancel{V}$ (m^3)	**n** (mol)	$\mathbf{E}_{int}$ (J)	–
B	673.15	1837.9	6.25×10^{-5}	$2.052\,5 \times 10^{-2}$	287.17	–
C	1023	2793.1	6.25×10^{-5}	$2.052\,5 \times 10^{-2}$	436.42	–
Process	**Q** (J)	$\mathbf{w}_{int}$ (J)	$\mathbf{\Delta E}_{int}$ (J)	$\mathbf{w}_{eng}$ (J)	$\mathbf{Q}_h$	Q_c
BC	149.25	0	149.25	**0**	149.25	0